

Exercise

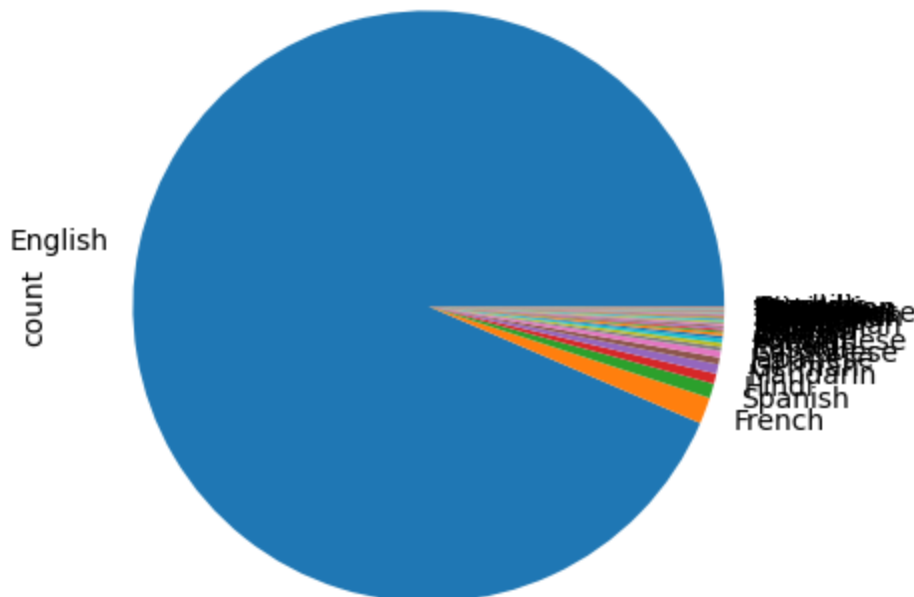
```
In [1]: import pandas as pd
import matplotlib.pyplot as plt
df=pd.read_csv('../data/movies.csv')
df.head()
```

```
Out[1]:
```

	color	director_name	num_critic_for_reviews	duration	director_facebook_li
0	Color	James Cameron	723.0	178.0	
1	Color	Gore Verbinski	302.0	169.0	56
2	Color	Sam Mendes	602.0	148.0	
3	Color	Christopher Nolan	813.0	164.0	2200
4	NaN	Doug Walker	NaN	NaN	13

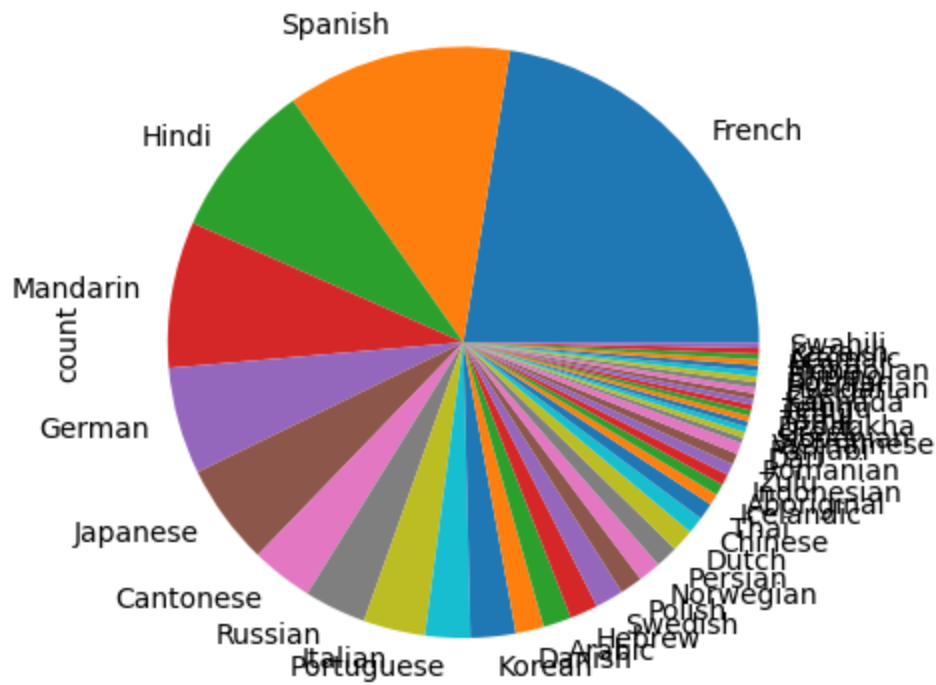
5 rows × 28 columns

```
In [14]: # find how many different languages do we have in this dataset
len(df.language.dropna().unique())
counts=df.language.dropna().value_counts()
counts.plot(kind='pie');
```



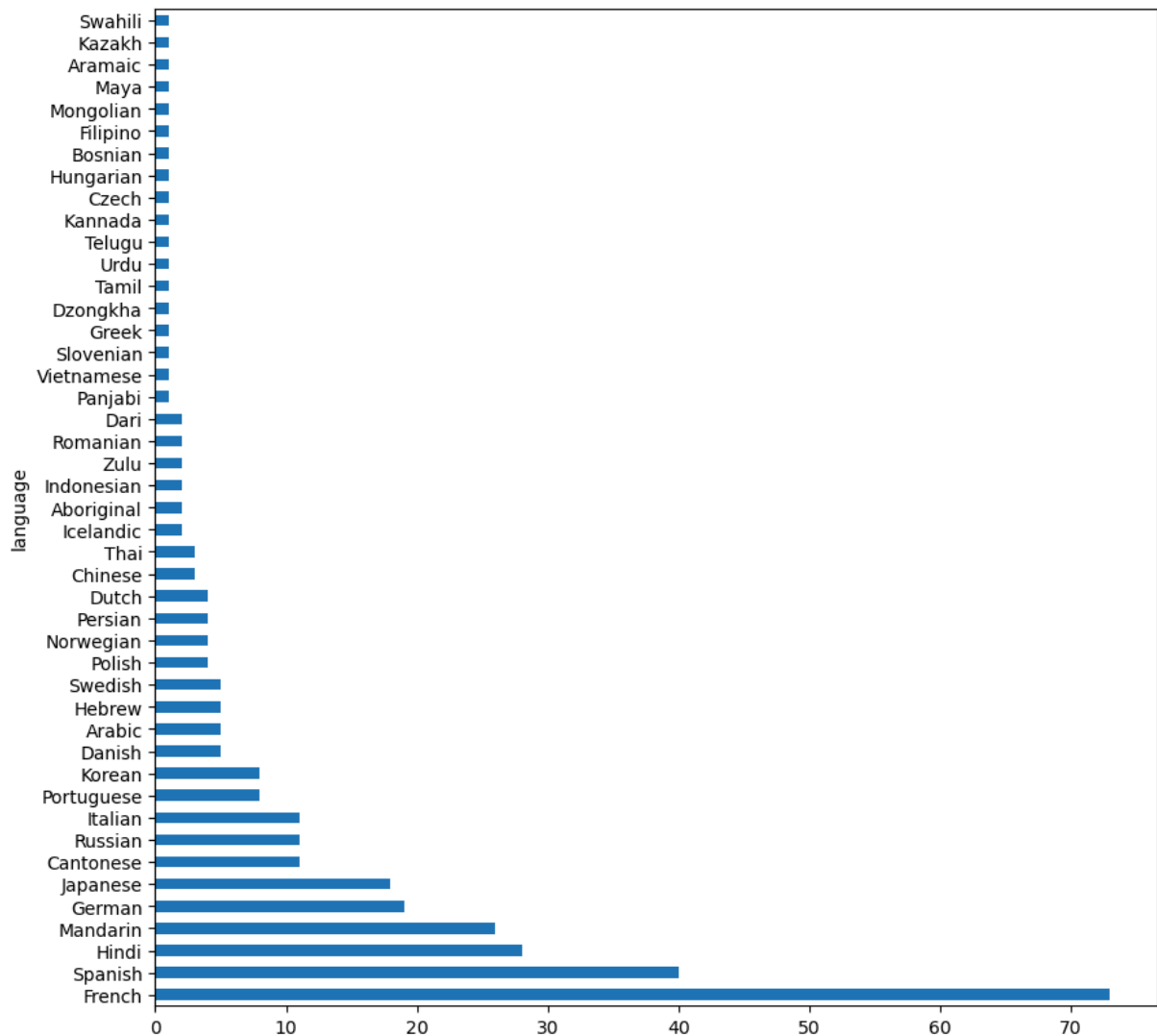
```
In [17]: counts[counts<100].plot(kind='pie')
```

Out[17]: <Axes: ylabel='count'>



```
In [20]: plt.figure(figsize=(10,10))
counts[counts<100].plot(kind='barh')
```

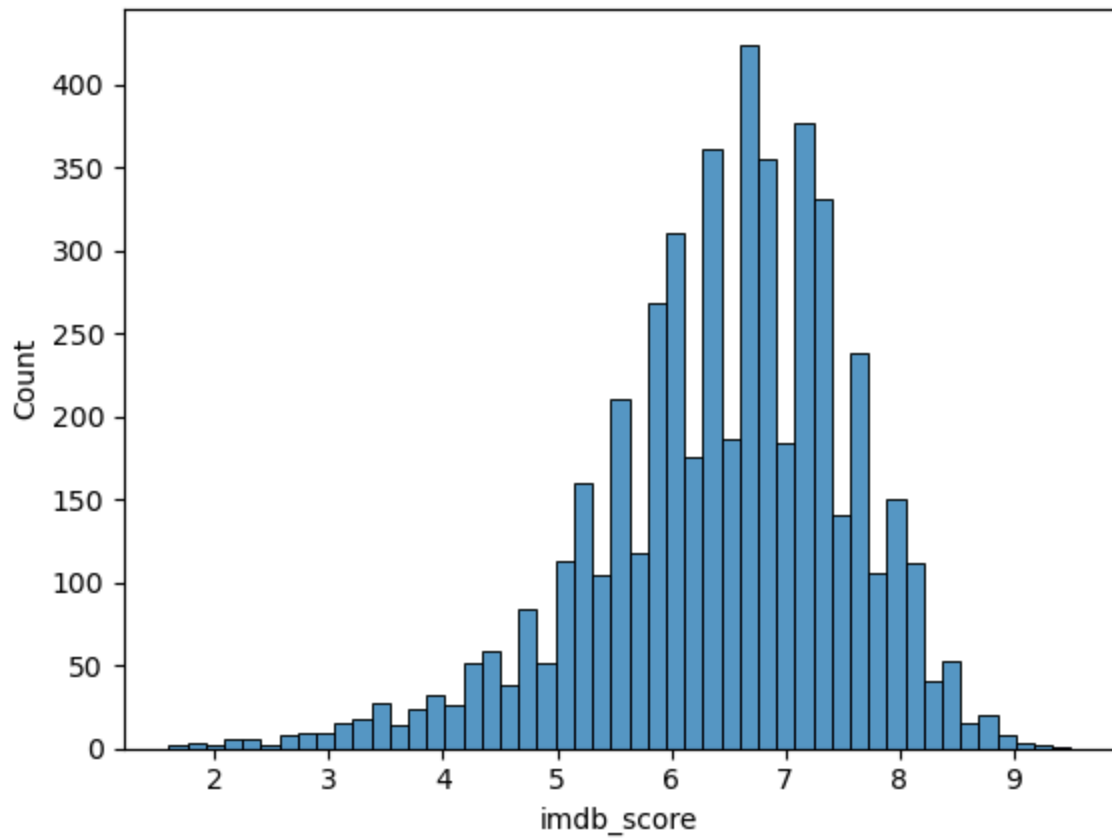
Out[20]: <Axes: ylabel='language'>



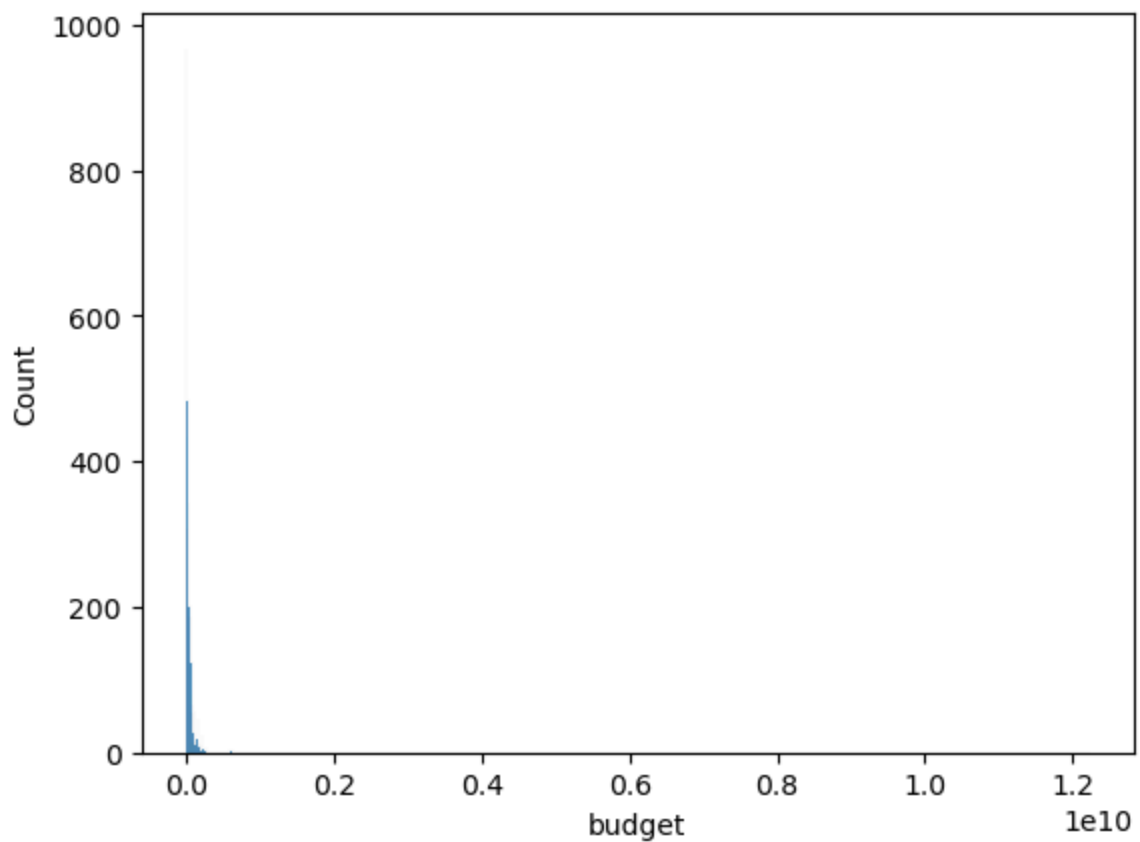
```
In [27]: # pip install seaborn
```

```
In [29]: # show how imdb_score is distributed in this dataset (show the histogram)
# plt.hist(df.imdb_score);
import seaborn as sns
sns.histplot(data=df, x='imdb_score')
```

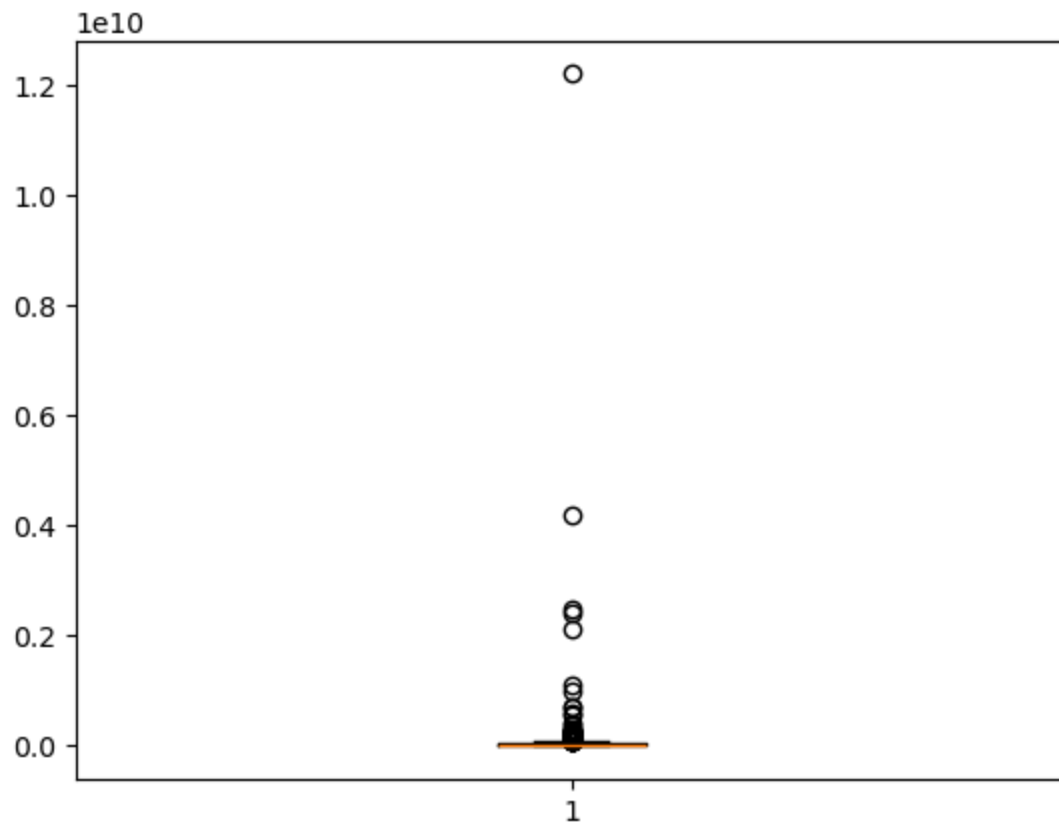
```
Out[29]: <Axes: xlabel='imdb_score', ylabel='Count'>
```



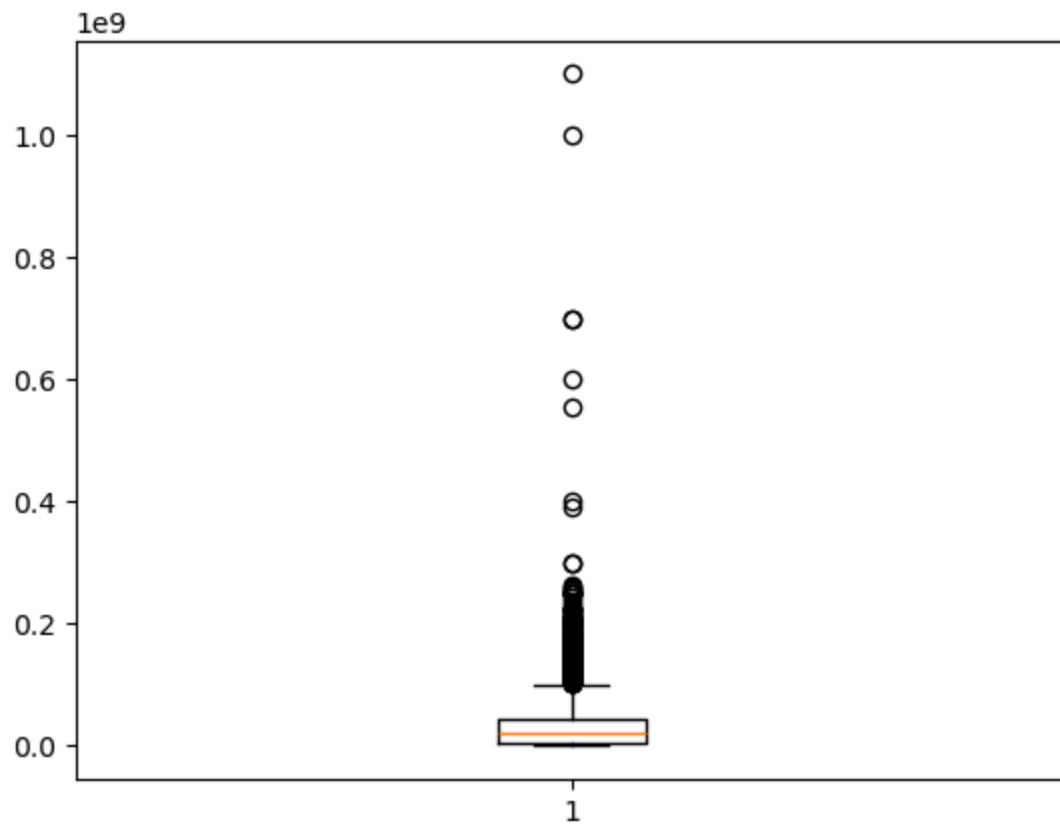
```
In [37]: # show the histogram of the movies budgets
# df.budget.dropna().plot(kind='hist');
sns.histplot(data=df, x='budget');
```



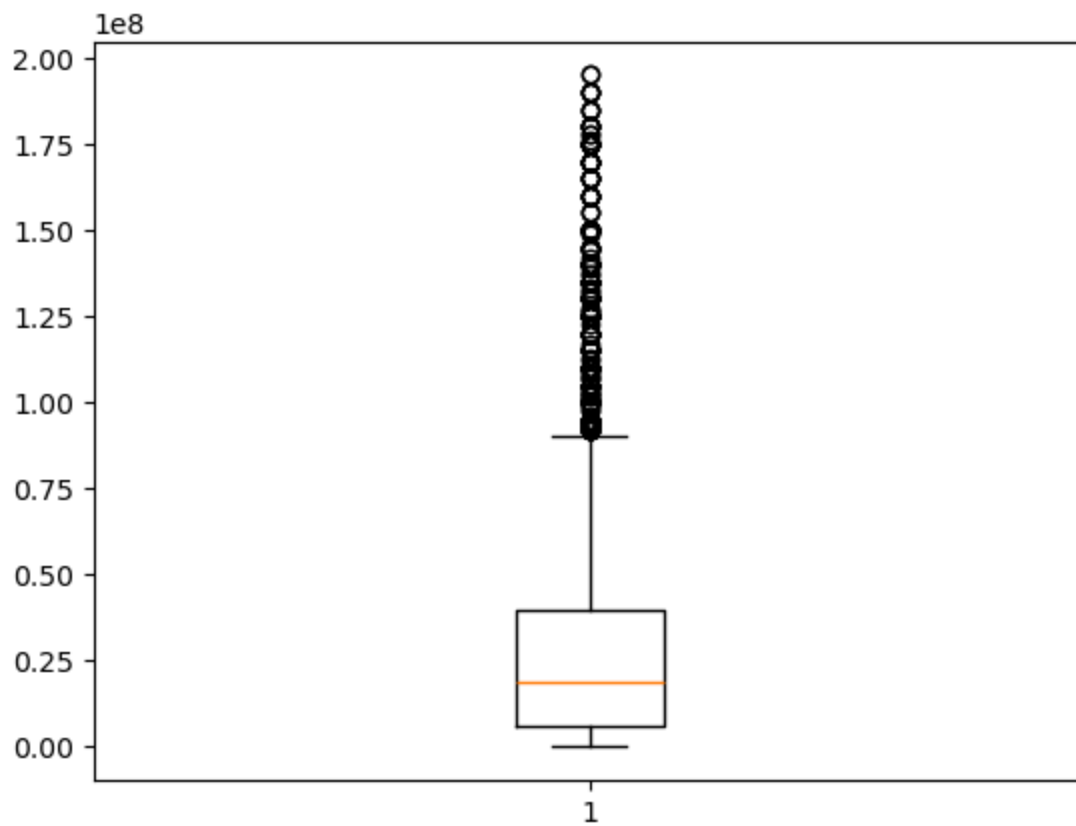
```
In [39]: plt.boxplot(df.budget.dropna());
```



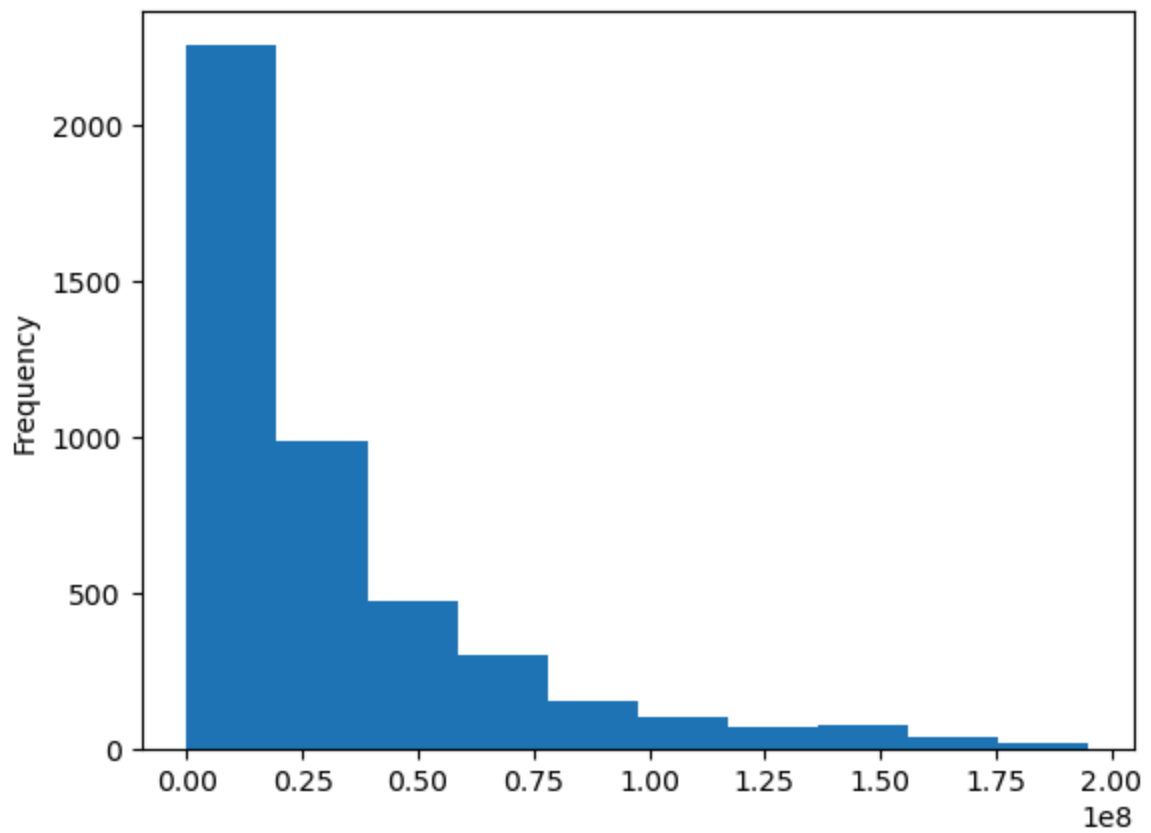
```
In [43]: budgets=df.budget.dropna()  
plt.boxplot(budgets[budgets<0.2e10]);
```



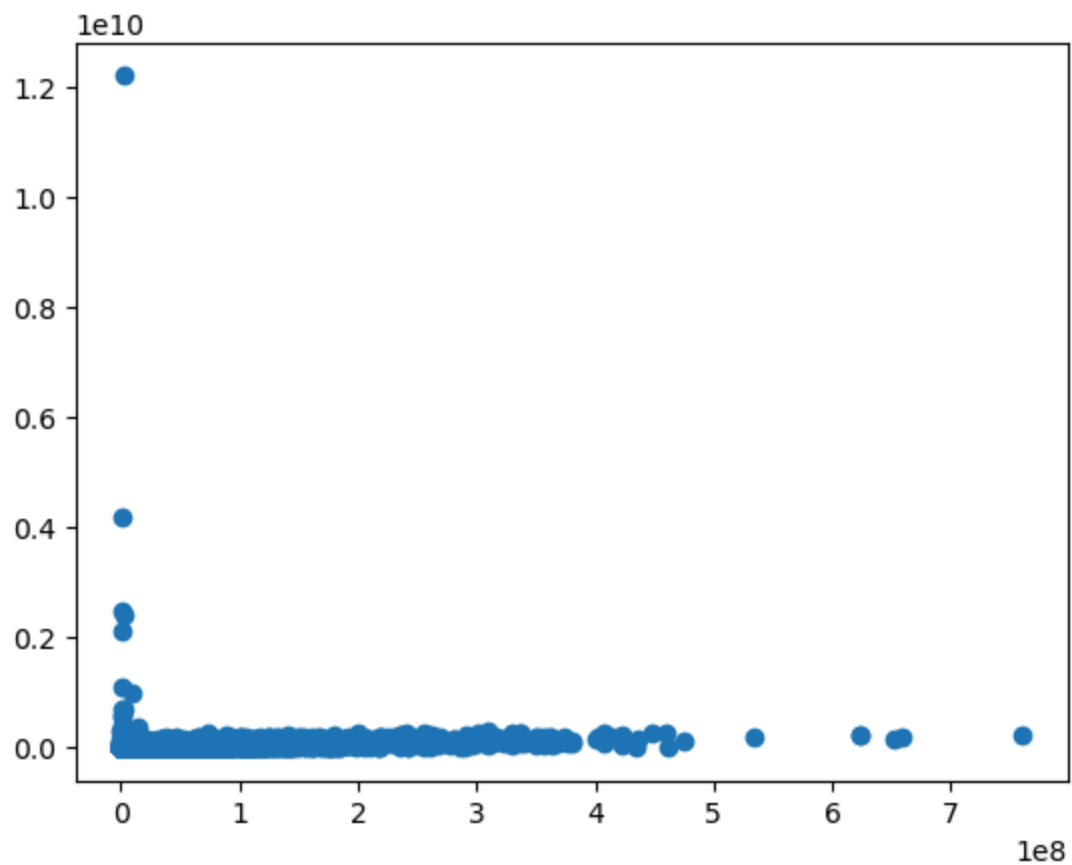
```
In [44]: plt.boxplot(budgets[budgets<0.2e9]);
```



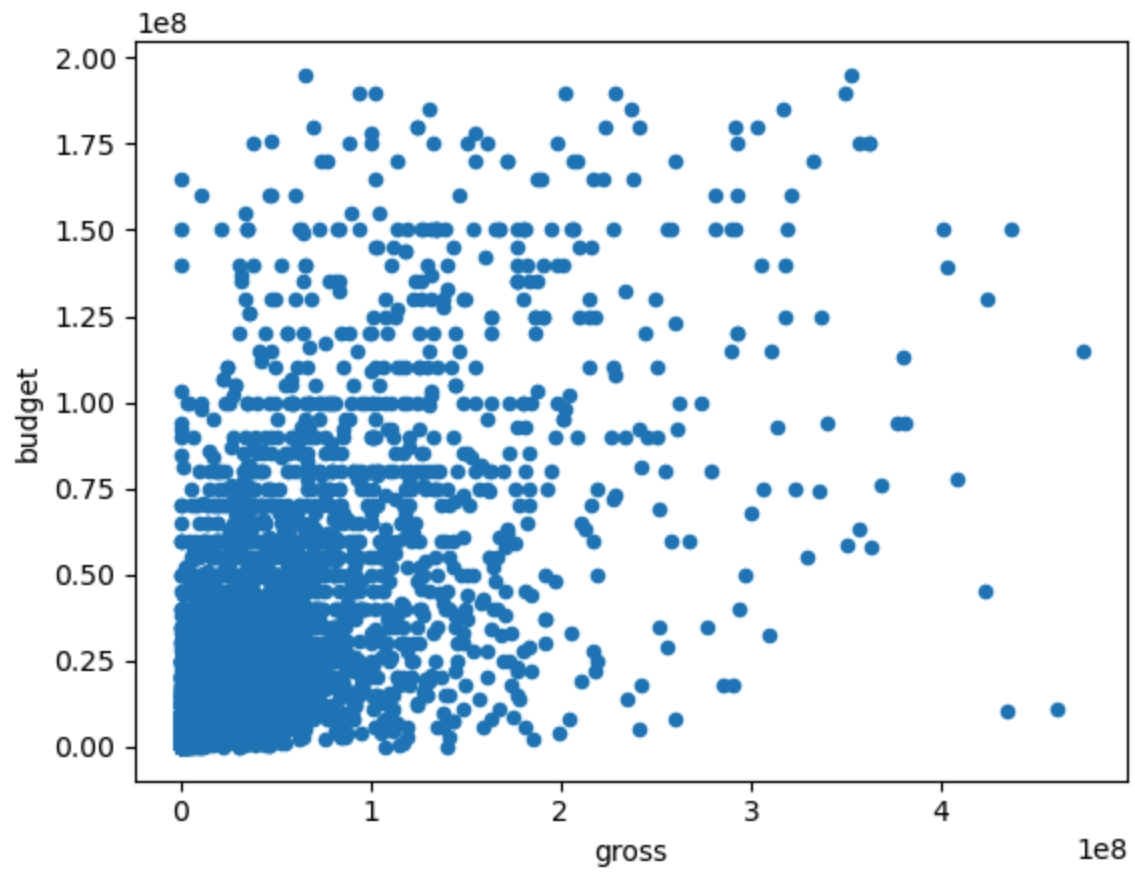
```
In [45]: budgets[budgets<0.2e9].plot(kind='hist');
```



```
In [46]: # Find out if there is a relation between gross income and the budget  
plt.scatter(df.gross, df.budget);
```



```
In [52]: df.loc[(df.gross<5e8)&(df.budget<0.2e9), ['gross', 'budget']].dropna().plot()
```



```
In [ ]:
```